NONMANDATORY APPENDIX O
ASSEMBLY BOLT STRESS DETERMINATION

O-1 INTRODUCTION

(22) **O-1.1 Scope**

This Appendix intends to provide guidance for the determination of an appropriate assembly bolt stress with due consideration for joint integrity. The detailed procedures provided in this Appendix are intended for flange joints for which controlled assembly methods are to be used. Provisions are made for two simple approaches and a joint component approach.

The historic use of a common single bolt stress across all flange sizes and ratings [e.g., 345 MPa (50 ksi)] can result in a gasket stress that does not provide sufficient margin to overcome the effects of creep/relaxation, pressure/external loads, and thermal loading. In addition, the use of this bolt stress can result in either loading bolts past their yield strength, as in the case of austenitic stainless steels, or loading flanges past their flange strength limit, causing permanent flange deformation. For this reason, the joint component approach outlined in this Appendix is preferred. However, for some sites, due to their limited range of operating conditions and flange configurations, the simpler gasket stress or single bolt stress approaches may offer sufficient joint integrity and simpler site execution.

The calculations contained in this Appendix should be used to assist in the selection of other aspects of joint assembly, such as the assembly method and whether additional steps, such as start-up retorque, are required. For example, the calculations may indicate that the required bolt load is close to the flange strength limit, which may require the use of more accurate assembly methods and/or the use of a start-up retorque to recover initial bolt load relaxation. A smaller range of bolt load between the minimum required and the maximum permissible indicates that greater care should be taken with assembly method selection, assembly procedure selection, and load control method.

O-1.2 Cautions

The provisions of this Appendix consider that the ASME PCC-1 guidelines for the joint component condition (flange surface finish, bolt spacing, flange rigidity, bolt condition, etc.) are within acceptable limits.

The methodology outlined in this Appendix assumes that the gaskets being used undergo a reasonable amount (>15%) of relaxation during the initial stages of operation, such that the effects of operational loads in increasing the bolt stress need not be considered (i.e., gasket relaxation will exceed any operational bolt-load increase). In some rare cases, this may not be the case, and the limits should then also be checked at both the ambient and operating bolt stress and temperatures. For most standard applications, this will not be necessary.

In addition, the methodology is for ductile materials (strain at tensile failure in excess of 15%). For brittle materials, the margin between the specified assembly bolt stress and the point of component failure may be considerably reduced and, therefore, additional safety factors should be introduced to guard against such failure.

The method does not consider the effect of fatigue, creep, or environmental damage mechanisms on either the bolt or flange. These additional modes of failure may also need to be considered for applications where they are found and additional reductions in assembly bolt stress may be required to avoid joint component failure.

O-1.3 Definitions (22)

A_b = bolt root area, mm^2 (in.^{2})
A_g = gasket area [$\pi/4$ ($G_{O.D.}{}^2 - G_{I.D.}{}^2$)], mm^2 (in.^{2})[1]
$G_{I.D.}$ = larger of the gasket sealing element or flange seating surface inner diameter, mm (in.)
$G_{O.D.}$ = smaller of the gasket sealing element or the flange seating surface outer diameter, mm (in.)
K = nut factor (for bolt material, lubricant/anti-seize, and temperature)
n_b = number of bolts
P_{max} = maximum design pressure, MPa (psi)
S_{ya} = flange yield stress at assembly, MPa (psi)
S_{yo} = flange yield stress at operation, MPa (psi)
Sb_{max} = maximum permissible bolt stress, MPa (psi)
Sb_{min} = minimum permissible bolt stress, MPa (psi)
Sb_{sel} = selected assembly bolt stress, MPa (psi)

[1] Where a gasket has additional gasket area, such as a pass partition gasket, which may not be as compressed as the main outer sealing element due to flange rotation, a reduced portion of that area, such as half the additional area, should be added to A_g.

Sf_{max} = maximum permissible bolt stress prior to flange damage, MPa (psi)
Sg_{max} = maximum permissible gasket stress, MPa (psi)
$Sg_{min\text{-}O}$ = minimum gasket operating stress, MPa (psi)
$Sg_{min\text{-}S}$ = minimum gasket seating stress, MPa (psi)
Sg_T = target assembly gasket stress, MPa (psi)
T_b = assembly bolt torque, N·m (ft-lb)
T_i = Target Torque Index based on a unit bolt stress, N·m/MPa (ft-lb/ksi)
θf_{max} = single flange rotation at Sf_{max}, deg
θg_{max} = maximum permissible single flange rotation for gasket at the maximum operating temperature, deg
ϕ_b = bolt diameter, mm (in.)
ϕ_g = fraction of gasket load remaining after relaxation

(22) O-2 ASSEMBLY BOLT STRESS SELECTION

It is recommended that bolt assembly stresses be established with due consideration of the following joint integrity issues:

(a) Sufficient Gasket Stress to Seal the Joint. The assembly bolt stress should provide sufficient gasket stress to seat the gasket and sufficient gasket stress during operation to maintain a seal.

(b) Damage to the Gasket. The assembly bolt stress should not be high enough to cause overcompression (physical damage) of the gasket or excessive rotation of the flange, which might lead to localized gasket overcompression or damage to the flange sealing surface.

(c) Damage to the Bolts. The specified bolt stress should be below the bolt yield point, such that bolt failure does not occur. In addition, the life of the bolt can be extended by specifying an even lower load. If hydraulic tensioners are used for assembly, then the appropriate load loss factors, per Nonmandatory Appendix Q, should be considered.

(d) Damage to the Flange. The assembly bolt stress should be selected such that permanent deformation of the flange does not occur. If the flange is deformed during assembly, it might leak during operation or successive assemblies might cause joint leakage due to excessive flange rotation. Leakage due to flange rotation may be due to the concentration of the gasket stress on the gasket outer diameter causing damage or additional relaxation. Another potential issue is the flange face outer diameter touching, which reduces the effective gasket stress. If hydraulic tensioners are used for assembly, then the pass B load loss factors, per Nonmandatory Appendix Q, should be considered against flange and gasket damage.

However, it is also important to consider the practicalities involved with the in-field application of the specified bolt stress. If a different assembly stress is specified for each flange in a plant, including all variations of standard piping flanges, then it is unlikely, without a significant assembly quality assurance plan, that success will be improved in the field by comparison to a simpler method. Depending on the complexity of the joints in a given plant, a simple approach (e.g., standard bolt stress per size across all standard flanges) may be more effective in preventing leakage than a more complex approach that includes consideration of the integrity of all joint components.

This Appendix outlines two approaches: the simpler single-assembly bolt stress approach (which is simpler to use but may result in damage to joint components) and a more complex joint component-based approach that considers the integrity of each component.

O-3 SIMPLE APPROACH

O-3.1 Required Information (22)

In order to determine a standard assembly bolt stress for a single joint or across all flanges, it is recommended that, as a minimum, the target gasket stress, Sg_T, for a given gasket type be considered. Further integrity issues, as outlined in section O-4 on the joint component approach, may also be considered, as deemed necessary.

O-3.2 Determining the Appropriate Bolt Stress (22)

The appropriate bolt stress for a range of typical joint configurations may be determined via eq. (O-1).

$$Sb_{sel} = Sg_T \frac{A_g}{n_b A_b} \tag{O-1}$$

The average bolt stress across the joints considered may then be selected and this value can be converted into a torque value using eq. (O-2M) for metric units or eq. (O-2) for U.S. Customary units.

$$T_b = Sb_{sel} K\, A_b \phi_b / 1\,000 \tag{O-2M}$$

$$T_b = Sb_{sel} K\, A_b \phi_b / 12 \tag{O-2}$$

As an alternative to these equations, Tables O-3.2-1M and O-3.2-1 provide reference tables that tabulate Target Torque Indices, T_i, based on eqs. (O-2M) and (O-2) using a unit bolt stress (e.g., substituting a value of 1 for Sb_{sel}), with nut factors of 0.15, 0.18, and 0.2. The final assembly bolt torque may then be obtained by using eq. (O-3). An example is shown in Notes (b)(2) and (b)(3) in Table O-3.2-1M and Notes (b)(2) and (b)(3) in Table O-3.2-1.

The nut factors provided in Tables O-3.2-1M and O-3.2-1 represent examples and may vary from actual values. Refer to Nonmandatory Appendix K for guidance in determining nut factors.

If another bolt stress or nut factor is required, then the table may be converted to the new values using eq. (O-3), where Sb'_{sel}, T'_b, and K' are the original values.

$$T_b = \frac{K}{K'}\frac{Sb_{\text{sel}}}{Sb'_{\text{sel}}}T'_b = T_i\frac{K}{K'}Sb_{\text{sel}} \tag{O-3}$$

O-4 JOINT COMPONENT APPROACH

(22) O-4.1 Required Information

There are several values that should be known prior to calculating the appropriate assembly bolt stress using the joint component approach.

(a) The maximum permissible flange rotation, $\theta g_{\max}$, at the assembly gasket stress and the gasket operating temperature should be obtained from industry test data or from the gasket manufacturer. There is presently no standard test for determining this value; however, typical limits vary from 0.3 deg for expanded PTFE gaskets to 1.0 deg for typical graphite-filled metallic gaskets (per flange). A suitable limit may be determined for a given site or application based on the calculation of the amount of rotation that presently exists in flanges in a given service using the gasket type in question, provided that rotation has not been associated with leakage.

(b) The maximum permissible bolt stress, $Sb_{\max}$, should be selected by the user. This value is intended to eliminate damage to the bolt or assembly equipment during assembly and may vary from site to site. It is typically in the range of 40% to 70% of ambient bolt yield stress (see section 10), and the bolt load is sufficiently high to prevent self-loosening.

(c) The minimum permissible bolt stress, $Sb_{\min}$, should be selected by the user. This value is intended to provide a lower limit such that bolting inaccuracies do not become a significant portion of the specified assembly bolt stress, Sb_{sel}, and the bolt load is sufficiently high to prevent self-loosening. The value is typically in the range of 140 MPa to 245 MPa (20 ksi to 35 ksi).

(d) The maximum permissible bolt stress for the flange, $Sf_{\max}$, should be determined, based on the particular flange configuration. This may be found using either elastic closed-form solutions or elastic–plastic finite element analysis, as outlined in section O-5. In addition, when the limits are being calculated, the flange rotation at that load, $\theta f_{\max}$, should also be determined. Example flange limit loads for elastic closed-form solutions and elastic–plastic finite element solutions are outlined in Tables O-4.1-1M through O-4.1-7.[2]

(e) The target assembly gasket stress, Sg_T, should be selected by the user considering user experience and industry test data or in consultation with the gasket manufacturer. The target gasket stress is based on the full gasket area and should be selected to be near the upper end of the acceptable gasket stress range, as this will give the most amount of buffer against joint leakage.

(f) The maximum assembly gasket stress, $Sg_{\max}$, should be obtained from industry test data or the gasket manufacturer. This value is the maximum compressive stress at the assembly temperature, based on the full gasket area, which the gasket can withstand without permanent damage (excessive leakage or lack of elastic recovery) to the gasket sealing element. Any value provided should include consideration of the effects of flange rotation for the type of flange being considered in increasing the gasket stress locally on the outer diameter.

(g) The minimum gasket seating stress, $Sg_{\text{min-S}}$, should be obtained from industry test data or the gasket manufacturer. This value is the minimum recommended compressive stress at the assembly temperature and is based on the full gasket area. The value is the stress that the gasket should be assembled to in order to obtain adequate redistribution of any filler materials into filler serrations and ensure an initial seal between the gasket and the flange faces.

(h) The minimum gasket operating stress, $Sg_{\text{min-O}}$, should be obtained from industry test data or the gasket manufacturer. This value is the minimum recommended compressive stress after offloading of the gasket by operational loads and is based on the full gasket area. This is the gasket stress that should be maintained on the gasket during operation in order to ensure that leakage does not occur.

(i) The gasket relaxation fraction, ϕ_g, should be obtained from industry test data or the gasket manufacturer for the gasket in flange assemblies of similar configuration (geometry, dimensions, and rigidity) to the ones being assessed. A default value of 0.7 may be used if data are not available.

O-4.2 Determining the Appropriate Bolt Stress (22)

Once the limits are defined, the following process may be used for each joint configuration. This process may be performed using a spreadsheet or software program, which allows the determination of many values simultaneously.

Step 1. Determine the target bolt stress in accordance with eq. (O-1).

Step 2. Determine if the bolt upper limit controls

$$Sb_{\text{sel}} = \text{min.}\ (Sb_{\text{sel}},\ Sb_{\max}) \tag{O-4}$$

Step 3. Determine if the bolt lower limit controls

$$Sb_{\text{sel}} = \text{max.}\ (Sb_{\text{sel}},\ Sb_{\min}) \tag{O-5}$$

[2] Limits are not presented for flanges less than NPS 2 (DN 50), as the consequence of gross plastic deformation of such small flanges is generally inconsequential to joint integrity. The smaller joint dimensions mean that the residual flange rotation must be significantly more severe when compared to a larger flange before it can be detected, let alone affect joint integrity. The joint component method may be applied to flange sizes or classes not listed in this Appendix or to small bore flanges using the method outlined in WRC Bulletin 538, but for small bore flanges the bolt load should not be excessively limited due to flange strength (i.e., minimum gasket stress levels should control the calculation, over flange strength).

Step 4. Determine if the flange limit controls[3]

$$Sb_{\text{sel}} = \text{min. } (Sb_{\text{sel}},\ Sf_{\max}) \tag{O-6}$$

Step 5. Check if the gasket assembly seating stress is achieved.

$$Sb_{\text{sel}} \geq Sg_{\text{min-}S}[A_g/(A_b n_b)] \tag{O-7}$$

Step 6. Check if the gasket operating stress is maintained.[4]

$$Sb_{\text{sel}} \geq \left[Sg_{\text{min-O}}\,A_g + \left(\pi/4\right)P_{\max}G_{\text{I.D.}}{}^2\right]/\left(\phi_g A_b n_b\right) \tag{O-8}$$

Step 7. Check if the gasket maximum stress is achieved.

$$Sb_{\text{sel}} \leq Sg_{\max}[A_g/(A_b n_b)] \tag{O-9}$$

Step 8. Check if the flange rotation limit is exceeded.

$$Sb_{\text{sel}} \leq Sf_{\max}(\theta g_{\max}/\theta f_{\max}) \tag{O-10}$$

If one of the final checks (Step 5 through Step 8) is exceeded, then judgment should be used to determine which controlling limit is more critical to the integrity and, therefore, what the selected bolt load ought to be. A table of assembly bolt torque values can then be calculated using eq. (O-2M) or eq. (O-2). An example table of assembly bolt stresses and torque values using this approach is outlined in Tables O-4.2-1 and O-4.2-2, respectively.

(22) O-4.3 Example Calculation

NPS 3 Class 300 Carbon Steel RFWN Flange Operating at Ambient Temperature (Identical Limits Used as Those in Table O-4-2.1) with a spiral-wound gasket per ASME B16.20 and nut factor per Table O-4.2-2

$$\begin{aligned}
A_b &= 0.3019\ \text{in.}^2 \\
A_b{\cdot}n_b &= 2.42\ \text{in.}^2 \\
A_g &= 5.17\ \text{in.}^2 \\
G_{\text{I.D.}} &= 4.00\ \text{in.} \\
n_b &= 8 \\
P_{\max} &= 750\ \text{psig (0.75 ksi)} \\
\phi_b &= 0.75\ \text{in.} \\
\phi_g &= 0.7
\end{aligned}$$

Determine Bolt Stress

Step 1.	Equation (O-1)	$Sb_{\text{sel}} = 30(5.17/2.42) = 64$ ksi
Step 2.	Equation (O-4)	$Sb_{\text{sel}} = \text{min. } (64,\ 75) = 64$ ksi
Step 3.	Equation (O-5)	$Sb_{\text{sel}} = \text{max. } (64,\ 35) = 64$ ksi
Step 4.	Table O-4.1-2	$Sf_{\max} = 63$ ksi (note: $S_{yo} = S_{ya}$)
Step 5.	Equation (O-6)	$Sb_{\text{sel}} = \text{min. } (64,\ 63) = 63$ ksi

Additional Checks

Step 6.	Equation (O-7)	$Sb_{\text{sel}} \geq 12.5\ (5.17/2.42) \geq 26.7$ ksi ✓
Step 7.	Equation (O-8)	$Sb_{\text{sel}} \geq [6.0 \times 5.17 + (\pi/4) \times 0.75 \times 4.00^2]/(0.7 \times 2.42) \geq 24$ ksi ✓
Step 8.	Equation (O-9)	$Sb_{\text{sel}} \leq 40\ (5.17/2.42) \leq 85$ ksi ✓
	Table O-4.1-4	$\theta f_{\max} = 0.32$ deg
	Equation (O-10)	$Sb_{\text{sel}} \leq 63\ (1.0/0.32) \leq 197$ ksi ✓
	Equation (O-2)	$T_b = 63{,}000 \times 0.2 \times 0.3019 \times 0.75/12$ $T_b \approx 240$ ft-lb
	Alternative: Use Table O-3.2-1	$T_b = 63$ ksi × 3.78 ft lb/ksi = 238 ft-lb

Note that for some flanges (e.g., NPS 8, class 150) the additional limits [eq. (O-7) through eq. (O-10)] are not satisfied. In those cases, engineering judgment should be used to determine which limits are more critical to the joint integrity, and the value of Sb_{sel} should be modified accordingly. It should be noted that the values presented are not hard limits (i.e., flange leakage will not occur if the gasket stress falls 0.1 psi below the limit) and therefore some leeway in using the values is to be considered normal.

O-5 DETERMINING FLANGE LIMITS

O-5.1 Elastic Analysis (22)

A series of elastic analysis limits have been determined that allow the calculation of the approximate assembly bolt stress that will cause significant permanent deformation of the flange. Since this bolt stress is approximate, and the flange specified minimum yield tends to be lower bound (i.e., the actual material yield will exceed the specified minimum yield strength), it is considered appropriate to use these limits without modification or additional safety factor. An explanation of the limits and equations used to determine the bolt stress can be found in WRC Bulletin 538. The inaccuracy of applied bolt stress may be considered in this process; however, caution is urged that including such considerations may lead to

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[3] In some cases (e.g., high-temperature stainless steel flanges), the yield strength of the flange may reduce significantly during operation. While this is important to consider, in some cases consideration of this effect will result in the selection of an assembly bolt load that will result in joint leakage. Once in operation, the bolt load becomes a secondary load (i.e., the load decreases with component yield). Therefore, the effect of temperature-driven component yield will be seen only as additional joint relaxation and minor permanent deformation. Often, the actual material yield for stainless steel is significantly above the specified minimum yield, and therefore it is considered more prudent to tighten to a higher load initially and risk possible permanent deformation than to tighten to a lower load and risk certain joint leakage. However, if the selected load minus bolt relaxation remains well above the flange strength at operating temperature, permanent deformation can eventually cause issues (such as not being able to insert the bolts due to residual flange rotation). In that case, it may be appropriate to use a higher level of analysis than that presented in this Appendix.

[4] Note that this simple treatment does not take into account the changes in bolt load during operation due to component elastic interaction. A more complex relationship for the operational gasket stress may be used in lieu of this equation that includes the effects of elastic interaction in changing the bolt stress. Note also that the use of the G term from ASME BPVC, Section VIII, Division 1, Mandatory Appendix 2 in place of $G_{\text{I.D.}}$ in this equation is considered acceptable.

the selection of bolt stress levels that risk leakage. For simplicity and to err on the side of a higher bolt load, scatter is not considered essential to include in this method.

O-5.2 Finite Element Analysis

A more accurate approach to determining the appropriate limit on assembly bolt load is to analyze the joint using elastic–plastic nonlinear finite element analysis (FEA). An explanation of the requirements for performing such an analysis is outlined in WRC Bulletin 538. It is not necessary to rerun the analysis for minor changes to the joint configuration (such as different gasket dimensions or minor changes to the flange material yield strength) as linear interpolation using the ratio of the change in gasket moment arm or ratio of the different yield strength can be used to estimate the assembly bolt stress limit for the new case.

Table O-3.2-1M
Reference Values (Target Torque Index) for Calculating Target Torque Values for Low-Alloy Steel Bolting Based on Unit Prestress of 1 MPa (Root Area) (Metric Series Threads) (22)

Basic Thread Designation	Target Torque Index, T_i, N·m/MPa (ft-lb/ksi), at Nut Factor, K, of		
	0.15	0.18	0.2
M12 × 1.75	0.13 (0.66)	0.16 (0.80)	0.17 (0.88)
M14 × 2	0.21 (1.07)	0.25 (1.28)	0.28 (1.42)
M16 × 2	0.33 (1.69)	0.40 (2.03)	0.44 (2.25)
M20 × 2.5	0.65 (3.31)	0.78 (3.97)	0.87 (4.42)
M22 × 2.5	0.90 (4.57)	1.08 (5.49)	1.20 (6.10)
M24 × 3	1.13 (5.73)	1.35 (6.87)	1.50 (7.63)
M27 × 3	1.68 (8.52)	2.01 (10.2)	2.23 (11.4)
M30 × 3.5	2.26 (11.5)	2.72 (13.8)	3.02 (15.3)
M33 × 3.5	3.11 (15.8)	3.74 (19.0)	4.15 (21.1)
M36 × 4	3.99 (20.3)	4.78 (24.3)	5.31 (27.0)
M39 × 4	5.20 (26.5)	6.24 (31.8)	6.94 (35.3)
M42 × 4.5	6.41 (32.6)	7.70 (39.1)	8.55 (43.5)
M45 × 4.5	8.07 (41.0)	9.68 (49.2)	10.8 (54.7)
M48 × 5	9.67 (49.2)	11.6 (59.0)	12.9 (65.6)
M52 × 5	12.6 (64.1)	15.1 (76.9)	16.8 (85.4)
M56 × 5.5	15.6 (79.6)	18.8 (95.5)	20.9 (106)
M64 × 6	23.7 (120)	28.4 (145)	31.6 (161)
M72 × 6	34.8 (177)	41.8 (212)	46.4 (236)
M80 × 6	48.9 (249)	58.7 (299)	65.2 (332)
M90 × 6	71.4 (363)	85.7 (436)	95.2 (484)
M100 × 6	99.8 (507)	120 (609)	133 (677)

GENERAL NOTES:

(a) The root areas are based on coarse-thread series for sizes M68 and smaller, and 6-mm pitch thread series for sizes M70 and larger. The determination of root area for metric threads is based on a 6g tolerance.

(b) There are many ways of calculating target torque values for bolted pressure joints. The basis for Target Torque Index values in this table are described below. When conditions vary from those considered in this table, such as different bolt materials or different coatings, refer to Nonmandatory Appendix K to compute appropriate torque values.

(1) The tabulated target torque values are based on working surfaces that comply with sections 4 and 8.

(2) The Target Torque Index, T_i, is determined from eq. (O-2), for the nut factor indicated and applying a unit bolt load of 1 MPa for Sb_{sel}. For example, for an M20 bolt using a nut factor of 0.2

$$T_i = Sb_{\text{sel}}\, K\, A_b\, \phi_b / 1\,000 = 1 \times 0.2 \times 217 \times 20/1\,000 = 0.87$$

(3) When the Target Torque Index, T_i, is determined, this value is multiplied by the selected assembly bolt load Sb_{sel} to arrive at the final torque value, T_b. For example, the M20 bolt in (2) is in the NPS 3 Class 300 flange described in the example calculation in para. O-4.3, for which the selected assembly bolt stress, Sb_{sel}, is 434 MPa. The target torque becomes

$$T_b = Sb_{\text{sel}} \times T_i = 434\ \text{MPa} \times 0.87\ \text{N·m/MPa} = 378\ \text{N·m}$$

This calculated value is rounded up to the nearest 5 N·m to become 380 N·m.

Table O-3.2-1

(22) **Reference Values (Target Torque Index) for Calculating Target Torque Values for Low-Alloy Steel Bolting Based on Unit Prestress of 1 ksi (Root Area) (Inch Series Threads)**

Nominal Bolt Size, in.	Target Torque Index, T_i, ft-lb/ksi (N·m/MPa), at Nut Factor, K, of 0.15	0.18	0.2
$\frac{1}{2}$	0.79 (0.15)	0.94 (0.19)	1.05 (0.21)
$\frac{5}{8}$	1.58 (0.31)	1.89 (0.37)	2.10 (0.41)
$\frac{3}{4}$	2.83 (0.56)	3.40 (0.67)	3.78 (0.74)
$\frac{7}{8}$	4.58 (0.90)	5.50 (1.08)	6.11 (1.20)
1	6.89 (1.35)	8.27 (1.63)	9.18 (1.81)
$1\frac{1}{8}$	10.2 (2.01)	12.3 (2.42)	13.7 (2.68)
$1\frac{1}{4}$	14.5 (2.85)	17.4 (3.43)	19.4 (3.81)
$1\frac{3}{8}$	19.9 (3.90)	23.8 (4.68)	26.5 (5.20)
$1\frac{1}{2}$	26.3 (5.18)	31.6 (6.22)	35.1 (6.91)
$1\frac{5}{8}$	34.1 (6.71)	41.0 (8.05)	45.5 (8.95)
$1\frac{3}{4}$	43.3 (8.52)	52.0 (10.2)	57.8 (11.4)
$1\frac{7}{8}$	53.9 (10.6)	64.7 (12.7)	71.9 (14.1)
2	66.3 (13.0)	79.5 (15.6)	88.3 (17.4)
$2\frac{1}{4}$	96.2 (18.9)	115 (22.7)	128 (25.2)
$2\frac{1}{2}$	134 (26.4)	161 (31.6)	179 (35.2)
$2\frac{3}{4}$	181 (35.6)	217 (42.7)	241 (47.4)
3	237 (46.6)	284 (55.9)	316 (62.1)
$3\frac{1}{4}$	304 (59.8)	365 (71.8)	406 (79.8)
$3\frac{1}{2}$	383 (75.3)	459 (90.3)	510 (100)
$3\frac{3}{4}$	474 (93.2)	569 (112)	632 (124)
4	579 (114)	694 (137)	771 (152)

GENERAL NOTES:

(a) The root areas are based on coarse-thread series for sizes 1 in. and smaller, and 8-pitch thread series for sizes $1\frac{1}{8}$ in. and larger. The determination of root area for inch series threads is based on a 2A tolerance.

(b) There are many ways of calculating target torque values for bolted pressure joints. The basis for Target Torque Index values in this table are described below. When conditions vary from those considered in this table, such as different bolt materials or different coatings, refer to Nonmandatory Appendix K to compute appropriate torque values.

(1) The tabulated target torque values are based on working surfaces that comply with sections 4 and 8.

(2) The Target Torque Index, T_i, is determined from eq. (O-2), for the nut factor indicated and applying a unit bolt load of 1 ksi for Sb_{sel}. For example, for a $\frac{3}{4}$ in. bolt using a nut factor of 0.2

$$T_i = Sb_{sel}\,K\,A_b\,\phi_b/12 = 1{,}000 \times 0.2 \times 0.302 \times 0.75/12 = 3.78$$

where the value of 1,000 represents 1,000 psi rather than 1 ksi to maintain consistency in units.

(3) When the Target Torque Index, T_i, is determined, this value is multiplied by the selected assembly bolt load Sb_{sel} to arrive at the final torque value, T_b. For example, the $\frac{3}{4}$ in. bolt in (2) is in the NPS 3 Class 300 flange described in the example calculation in para. O-4.3, for which the selected assembly bolt stress, Sb_{sel}, is 63 ksi. The target torque becomes

$$T_b = Sb_{sel} \times T_i = 63\text{ ksi} \times 3.78\text{ ft-lb/ksi} = 238\text{ ft-lb}$$

This calculated value is rounded up to the nearest 5 ft-lb to become 240 ft-lb.

Table O-4.1-1M
Pipe Wall Thickness Used for Following Tables (mm)

NPS	Class 150	300	600	900	1500	2500
2	1.65	1.65	3.91	2.77	5.54	8.74
$2\frac{1}{2}$	2.11	2.11	3.05	5.16	7.01	14.02
3	2.11	2.11	3.05	5.49	7.62	15.24
4	2.11	2.11	6.02	6.02	11.13	17.12
5	2.77	2.77	6.55	9.52	12.70	19.05
6	2.77	2.77	7.11	10.97	14.27	23.12
8	2.77	3.76	8.18	12.70	20.62	30.10
10	3.40	7.80	12.70	15.09	25.40	37.49
12	3.96	8.38	12.70	17.48	28.58	44.47
14	3.96	6.35	12.70	19.05	31.75	...
16	4.19	7.92	14.27	23.83	34.93	...
18	4.78	9.53	20.62	26.19	44.45	...
20	4.78	9.53	20.62	32.54	44.45	...
24	5.54	14.27	24.61	38.89	52.37	...
26	7.92	12.70	23.73	35.09	...	...
28	7.92	12.70	25.56	37.79	...	...
30	6.35	15.88	27.38	40.49	...	...
32	7.92	15.88	29.21	43.19	...	...
34	7.92	15.88	31.03	45.89	...	...
36	7.92	19.05	32.86	48.59	...	...
38	9.53	17.60	34.69	51.29	...	...
40	9.53	18.52	36.51	53.99	...	...
42	9.53	19.45	38.34	56.69	...	...
44	9.53	20.37	40.16	59.39	...	...
46	9.53	21.30	41.99	62.09	...	...
48	9.53	22.23	43.81	64.79	...	...

Table O-4.1-1
Pipe Wall Thickness Used for Following Tables (in.)

NPS	Class 150	300	600	900	1500	2500
2	0.065	0.065	0.154	0.109	0.218	0.344
$2\frac{1}{2}$	0.083	0.083	0.120	0.203	0.276	0.552
3	0.083	0.083	0.120	0.216	0.300	0.600
4	0.083	0.083	0.237	0.237	0.438	0.674
5	0.109	0.109	0.258	0.375	0.500	0.750
6	0.109	0.109	0.280	0.432	0.562	0.910
8	0.109	0.148	0.322	0.500	0.812	1.185
10	0.134	0.307	0.500	0.594	1.000	1.476
12	0.156	0.330	0.500	0.688	1.125	1.751
14	0.156	0.250	0.500	0.750	1.250	...
16	0.165	0.312	0.562	0.938	1.375	...
18	0.188	0.375	0.812	1.031	1.750	...
20	0.188	0.375	0.812	1.281	1.750	...
24	0.218	0.562	0.969	1.531	2.062	...
26	0.312	0.500	0.934	1.382	...	...
28	0.312	0.500	1.006	1.488	...	...
30	0.250	0.625	1.078	1.594	...	...
32	0.312	0.625	1.150	1.700	...	...
34	0.312	0.625	1.222	1.807	...	...
36	0.312	0.750	1.294	1.913	...	...
38	0.375	0.693	1.366	2.019	...	...
40	0.375	0.729	1.437	2.126	...	...
42	0.375	0.766	1.509	2.232	...	...
44	0.375	0.802	1.581	2.338	...	...
46	0.375	0.839	1.653	2.444	...	...
48	0.375	0.875	1.725	2.551	...	...

Table O-4.1-2M Bolt Stress Limit for SA-105 Steel Flanges Using Elastic–Plastic FEA (MPa)

ASME B16.5 and ASME B16.47 Series A — Weld Neck						
	Class					
NPS	150	300	600	900	1500	2500
2	579	398	579	434	471	471
2½	688	326	434	398	471	543
3	724	434	615	579	471	579
4	543	615	688	434	507	507
5	543	724	652	507	543	543
6	724	579	579	579	615	579
8	724	579	615	507	579	579
10	579	543	543	507	615	579
12	724	543	507	543	579	615
14	579	434	471	543	543	...
16	543	434	471	579	507	...
18	724	471	579	543	543	...
20	615	507	507	579	507	...
24	615	471	507	543	507	...
26	253	253	362	434	...	...
28	217	253	326	398	...	...
30	253	290	434	434	...	...
32	217	253	398	434	...	...
34	190	290	434	398	...	...
36	217	253	398	434	...	...
38	253	579	579	543	...	...
40	217	543	615	543	...	...
42	253	543	615	579	...	...
44	226	579	615	543	...	...
46	253	615	652	543	...	...
48	253	507	579	579	...	...

Table O-4.1-2 Bolt Stress Limit for SA-105 Steel Flanges Using Elastic–Plastic FEA (ksi)

ASME B16.5 and ASME B16.47 Series A — Weld Neck						
	Class					
NPS	150	300	600	900	1500	2500
2	84	58	84	63	68	68
2½	100	47	63	58	68	79
3	105	63	89	84	68	84
4	79	89	100	63	74	74
5	79	105	95	74	79	79
6	105	84	84	84	89	84
8	105	84	89	74	84	84
10	84	79	79	74	89	84
12	105	79	74	79	84	89
14	84	63	68	79	79	...
16	79	63	68	84	74	...
18	105	68	84	79	79	...
20	89	74	74	84	74	...
24	89	68	74	79	74	...
26	37	37	53	63	...	...
28	32	37	47	58	...	...
30	37	42	63	63	...	...
32	32	37	58	63	...	...
34	28	42	63	58	...	...
36	32	37	58	63	...	...
38	37	84	84	79	...	...
40	32	79	89	79	...	...
42	37	79	89	84	...	...
44	33	84	89	79	...	...
46	37	89	95	79	...	...
48	37	74	84	84	...	...

Table O-4.1-3
Flange Rotation for SA-105 Steel Flanges Loaded to Table O-4.1-2M/Table O-4.1-2 Bolt Stress Using Elastic–Plastic FEA (deg)

ASME B16.5 and ASME B16.47 Series A — Weld Neck						
	Class					
NPS	150	300	600	900	1500	2500
2	0.37	0.34	0.23	0.21	0.20	0.16
2½	0.36	0.31	0.24	0.20	0.21	0.17
3	0.23	0.32	0.26	0.26	0.22	0.16
4	0.50	0.37	0.29	0.26	0.21	0.17
5	0.56	0.33	0.29	0.28	0.20	0.17
6	0.61	0.41	0.30	0.27	0.21	0.16
8	0.46	0.45	0.31	0.28	0.21	0.17
10	0.70	0.43	0.34	0.30	0.21	0.17
12	0.74	0.48	0.35	0.34	0.22	0.16
14	0.68	0.48	0.39	0.33	0.24	...
16	0.83	0.48	0.39	0.34	0.23	...
18	0.88	0.51	0.41	0.33	0.24	...
20	0.87	0.58	0.40	0.32	0.24	...
24	0.95	0.59	0.41	0.31	0.26	...
26	0.87	0.59	0.43	0.35	...	...
28	0.84	0.50	0.40	0.37	...	...
30	0.97	0.60	0.43	0.35	...	...
32	0.98	0.49	0.48	0.37	...	...
34	0.87	0.52	0.41	0.35	...	...
36	0.85	0.51	0.44	0.38	...	...
38	1.09	0.51	0.39	0.34	...	...
40	0.93	0.52	0.43	0.37	...	...
42	1.04	0.60	0.43	0.35	...	...
44	0.91	0.54	0.43	0.35	...	...
46	1.00	0.52	0.43	0.37	...	...
48	1.04	0.63	0.42	0.35	...	...

Table O-4.1-4M Bolt Stress Limit for SA-105 Steel Flanges Using Elastic Closed Form Analysis (MPa)

ASME B16.5 and ASME B16.47 Series A — Weld Neck

	Class					
NPS	150	300	600	900	1500	2500
2	450	310	515	332	413	447
2½	576	284	388	377	441	496
3	724	394	545	517	432	531
4	445	561	633	417	492	454
5	402	724	663	468	528	501
6	541	593	630	543	605	535
8	724	614	657	463	576	557
10	503	639	566	444	627	543
12	712	607	563	494	554	594
14	583	454	513	526	485	...
16	563	398	508	532	487	...
18	614	472	594	534	521	...
20	568	451	482	545	501	...
24	479	365	450	546	481	...
26	218	242	359	448	...	...
28	193	264	354	399	...	...
30	228	290	447	465	...	...
32	173	272	396	460	...	...
34	160	296	463	418	...	...
36	207	261	404	436	...	...
38	211	557	623	551	...	...
40	199	536	634	532	...	...
42	218	581	626	585	...	...
44	221	676	638	570	...	...
46	238	724	687	563	...	...
48	222	524	605	625	...	...

ASME B16.5 — Slip-On

	Class				
NPS	150	300	600	900	1500
2	724	360	572	423	413
2½	534	321	410	377	441
3	714	446	563	518	...
4	394	594	601	467	...
5	446	678	507	492	...
6	603	458	495	536	...
8	724	538	515	456	...
10	477	472	430	429	...
12	674	476	421	468	...
14	445	283	344	504	...
16	453	320	370	509	...
18	561	376	546	514	...
20	487	428	499	524	...
24	535	395	500	528	...

Table O-4.1-4 Bolt Stress Limit for SA-105 Steel Flanges Using Elastic Closed Form Analysis (ksi)

ASME B16.5 and ASME B16.47 Series A — Weld Neck

	Class					
NPS	150	300	600	900	1500	2500
2	65	45	75	48	60	65
2½	83	41	56	55	64	72
3	105	57	79	75	63	77
4	65	81	92	61	71	66
5	58	105	96	68	77	73
6	78	86	91	79	88	78
8	105	89	95	67	83	81
10	73	93	82	64	91	79
12	103	88	82	72	80	86
14	84	66	74	76	70	...
16	82	58	74	77	71	...
18	89	69	86	77	76	...
20	82	65	70	79	73	...
24	69	53	65	79	70	...
26	32	35	52	65	...	...
28	28	38	51	58	...	...
30	33	42	65	67	...	...
32	25	40	58	67	...	...
34	23	43	67	61	...	...
36	30	38	59	63	...	...
38	31	81	90	80	...	...
40	29	78	92	77	...	...
42	32	84	91	85	...	...
44	32	98	93	83	...	...
46	35	105	100	82	...	...
48	32	76	88	91	...	...

ASME B16.5 — Slip-On

	Class				
NPS	150	300	600	900	1500
2	105	52	83	61	60
2½	77	47	60	55	64
3	103	65	82	75	...
4	57	86	87	68	...
5	65	98	74	71	...
6	87	66	72	78	...
8	105	78	75	66	...
10	69	68	62	62	...
12	98	69	61	68	...
14	65	41	50	73	...
16	66	46	54	74	...
18	81	55	79	75	...
20	71	62	72	76	...
24	78	57	73	77	...

Table O-4.1-5
Flange Rotation for SA-105 Steel Flanges Loaded to Table O-4.1-4M/Table O-4.1-4 Bolt Stress Using Elastic Closed Form Analysis (deg)

ASME B16.5 and ASME B16.47 Series A — Weld Neck						
	Class					
NPS	150	300	600	900	1500	2500
2	0.20	0.20	0.15	0.13	0.09	0.08
2½	0.22	0.19	0.17	0.11	0.09	0.07
3	0.20	0.22	0.19	0.15	0.12	0.08
4	0.28	0.27	0.19	0.17	0.14	0.10
5	0.29	0.26	0.20	0.18	0.14	0.10
6	0.33	0.32	0.24	0.16	0.15	0.10
8	0.35	0.36	0.28	0.18	0.15	0.11
10	0.44	0.40	0.27	0.17	0.16	0.10
12	0.46	0.42	0.32	0.21	0.15	0.11
14	0.46	0.38	0.35	0.24	0.15	...
16	0.54	0.36	0.36	0.23	0.17	...
18	0.54	0.41	0.34	0.26	0.18	...
20	0.60	0.39	0.33	0.24	0.19	...
24	0.59	0.37	0.34	0.26	0.20	...
26	0.77	0.55	0.42	0.33	...	...
28	0.79	0.56	0.43	0.33	...	...
30	0.88	0.58	0.42	0.34	...	...
32	0.84	0.58	0.43	0.34	...	...
34	0.85	0.57	0.43	0.34	...	...
36	0.90	0.56	0.43	0.34	...	...
38	0.93	0.71	0.48	0.35	...	...
40	0.93	0.71	0.48	0.35	...	...
42	0.94	0.71	0.48	0.36	...	...
44	0.96	0.71	0.48	0.36	...	...
46	0.98	0.71	0.48	0.36	...	...
48	0.95	0.71	0.48	0.36	...	...

Table O-4.1-5
Flange Rotation for SA-105 Steel Flanges Loaded to Table O-4.1-4M/Table O-4.1-4 Bolt Stress Using Elastic Closed Form Analysis (deg) (Cont'd)

ASME B16.5 — Slip-On					
	Class				
NPS	150	300	600	900	1500
2	0.34	0.28	0.21	0.14	0.10
2½	0.35	0.29	0.24	0.12	0.10
3	0.40	0.32	0.27	0.21	...
4	0.52	0.38	0.27	0.21	...
5	0.64	0.43	0.30	0.20	...
6	0.73	0.49	0.33	0.20	...
8	0.84	0.57	0.38	0.22	...
10	1.02	0.59	0.40	0.27	...
12	1.09	0.66	0.47	0.33	...
14	1.14	0.70	0.50	0.33	...
16	1.26	0.76	0.52	0.33	...
18	1.34	0.80	0.52	0.34	...
20	1.38	0.86	0.55	0.33	...
24	1.52	0.91	0.58	0.33	...

Table O-4.1-6M
Bolt Stress Limit for SA-182 F304 Steel Flanges Using Elastic–Plastic FEA (MPa)

ASME B16.5 and ASME B16.47 Series A — Weld Neck						
	Class					
NPS	150	300	600	900	1500	2500
2	434	326	471	362	362	398
2½	543	253	362	326	398	434
3	724	362	507	471	362	471
4	362	471	543	362	434	434
5	398	615	543	398	434	434
6	543	471	471	471	471	471
8	579	471	507	434	471	471
10	434	434	434	434	507	471
12	471	434	434	434	471	507
14	434	326	362	434	434	...
16	362	362	362	471	434	...
18	398	398	471	471	434	...
20	362	398	398	471	434	...
24	362	362	398	434	398	...
26	217	181	290	362	...	...
28	181	217	290	326	...	...
30	217	217	362	362	...	...
32	181	217	326	362	...	...
34	172	253	362	326	...	...
36	181	217	326	362	...	...
38	181	471	471	434	...	...
40	145	434	507	434	...	...
42	217	434	507	471	...	...
44	154	471	471	434	...	...
46	217	507	507	434	...	...
48	217	398	471	471	...	...

Table O-4.1-6
Bolt Stress Limit for SA-182 F304 Steel Flanges Using Elastic–Plastic FEA (ksi)

ASME B16.5 and ASME B16.47 Series A — Weld Neck						
	Class					
NPS	150	300	600	900	1500	2500
2	63	47	68	53	53	58
2½	79	37	53	47	58	63
3	105	53	74	68	53	68
4	53	68	79	53	63	63
5	58	89	79	58	63	63
6	79	68	68	68	68	68
8	84	68	74	63	68	68
10	63	63	63	63	74	68
12	68	63	63	63	68	74
14	63	47	53	63	63	...
16	53	53	53	68	63	...
18	58	58	68	68	63	...
20	53	58	58	68	63	...
24	53	53	58	63	58	...
26	32	26	42	53	...	...
28	26	32	42	47	...	...
30	32	32	53	53	...	...
32	26	32	47	53	...	...
34	25	37	53	47	...	...
36	26	32	47	53	...	...
38	26	68	68	63	...	...
40	21	63	74	63	...	...
42	32	63	74	68	...	...
44	22	68	68	63	...	...
46	32	74	74	63	...	...
48	32	58	68	68	...	...

Table O-4.1-7
Flange Rotation for SA-182 F304 Steel Flanges Loaded to Table O-4.1-6M/Table O-4.1-6 Bolt Stress Using Elastic–Plastic FEA (deg)

ASME B16.5 and ASME B16.47 Series A — Weld Neck						
	Class					
NPS	150	300	600	900	1500	2500
2	0.47	0.34	0.21	0.17	0.15	0.16
$2^1/_2$	0.40	0.29	0.20	0.20	0.24	0.13
3	0.21	0.27	0.29	0.23	0.16	0.12
4	0.55	0.41	0.25	0.21	0.19	0.15
5	0.61	0.32	0.27	0.25	0.20	0.18
6	0.64	0.38	0.27	0.24	0.17	0.15
8	0.46	0.42	0.34	0.25	0.19	0.15
10	0.91	0.47	0.26	0.26	0.17	0.15
12	0.79	0.37	0.31	0.26	0.20	0.17
14	0.89	0.41	0.28	0.25	0.19	...
16	1.02	0.41	0.29	0.31	0.20	...
18	0.93	0.54	0.28	0.25	0.18	...
20	1.02	0.53	0.35	0.29	0.20	...
24	1.12	0.44	0.37	0.24	0.23	...
26	0.81	0.53	0.33	0.29	...	...
28	0.52	0.45	0.37	0.25	...	...
30	0.91	0.41	0.35	0.29	...	...
32	0.59	0.43	0.31	0.31	...	...
34	0.68	0.37	0.34	0.24	...	...
36	0.54	0.44	0.30	0.31	...	...
38	1.00	0.46	0.35	0.26	...	...
40	0.91	0.55	0.35	0.28	...	...
42	0.52	0.64	0.34	0.32	...	...
44	0.54	0.48	0.34	0.27	...	...
46	1.00	0.55	0.41	0.28	...	...
48	0.51	0.55	0.38	0.32	...	...

Table O-4.2-1
Example Bolt Stress for SA-105 Steel Weld-Neck Flanges, SA-193 B7 Steel Bolts, and Spiral-Wound Gasket With Inner Ring (ksi)
ASME B16.5 and ASME B16.47 Series A — Weld Neck

	Calculated Bolt Stress (ksi)					
NPS	150	300	600	900	1500	2500
2	75	56	56	43	43	35
$2^1/_2$	75	44	44	40	40	35
3	75	63	64	60	38	35
4	75	75	75	49	42	35
5	75	75	75	48	36	35
6	75	75	74	56	38	35
8	75	75	75	44	35	35
10	75	75	62	40	35	35
12	75	75	66	49	35	35
14	75	63	54	44	35	
16	75	63	58	42	35	
18	75	68	62	45	35	
20	75	74	57	38	35	
24	75	68	50	35	35	
26	37	37	41	35		
28	35	37	38	35		
30	37	42	41	35		
32	35	37	35	35		
34	35	42	36	35		
36	35	37	35	35		
38	37	75	39	35		
40	35	68	36	35		
42	37	71	35	35		
44	35	68	35	35		
46	37	69	35	35		
48	37	63	35	35		

Legend:

= limited by min. bolt stress

= limited by max. bolt stress

= limited by max. gasket stress

= limited by max. flange stress

GENERAL NOTE: Example limits used in the analysis:

Sb_{min} = 35 ksi
Sb_{max} = 75 ksi
Sf_{max} = from Table O-4.1-2
Sg_t = 30 ksi
Sg_{max} = 40 ksi
$Sg_{min\text{-}S}$ = 12.5 ksi
$Sg_{min\text{-}O}$ = 6 ksi
θg_{max} = 1.0 deg

Table O-4.2-2 Example Assembly Bolt Torque for SA-105 Steel Weld-Neck Flanges, SA-193 B7 Steel Bolts, and Spiral-Wound Gasket With Inner Ring (ft-lb)

ASME B16.5 and ASME B16.47 Series A — Weld Neck						
	Class					
NPS	150	300	600	900	1500	2500
2	160	120	120	265	265	325
$2\frac{1}{2}$	160	170	170	370	370	480
3	160	240	245	370	520	680
4	160	285	460	665	810	1,230
5	285	285	690	935	1,275	2,025
6	285	285	680	765	1,015	3,095
8	285	460	1,025	1,175	1,615	3,095
10	460	690	1,210	1,070	2,520	6,260
12	460	1,025	1,270	1,290	3,095	8,435
14	690	860	1,430	1,540	4,495	...
16	690	1,220	2,035	1,915	6,260	...
18	1,025	1,325	2,825	3,210	8,435	...
20	1,025	1,425	2,590	3,380	11,070	...
24	1,455	2,400	3,570	6,260	17,865	...
26	715	1,675	2,950	8,435	...	...
28	680	1,675	3,370	11,070	...	...
30	715	2,425	3,620	11,070	...	...
32	1,230	2,645	4,495	14,195	...	...
34	1,230	3,025	4,610	17,865	...	...
36	1,230	3,250	6,260	17,865	...	...
38	1,295	2,635	5,050	17,865	...	...
40	1,230	3,085	4,670	17,865	...	...
42	1,295	3,230	6,260	17,865	...	...
44	1,230	3,910	6,260	22,120	...	...
46	1,295	4,985	6,260	27,000	...	...
48	1,295	4,570	8,435	27,000	...	...

GENERAL NOTES:
(a) Nut factor used: K = 0.2.
(b) Torque rounded up to nearest 5 ft-lb.